

CS 171 / EE 142: Intro to ML and DM

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Slide Set 12: Feature Normalization



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
- DO NOT REDISTRIBUTE
 - ▶ For use only by enrolled students in the course

Feature Scaling

- Distance metric learning algorithms (beyond the scope of this course)

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$$b_j = \max_i x_{i,j}$$

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- Scale and translate so that the mean is 0 and the standard deviation is 1 (z-score normalization or z-normalization)

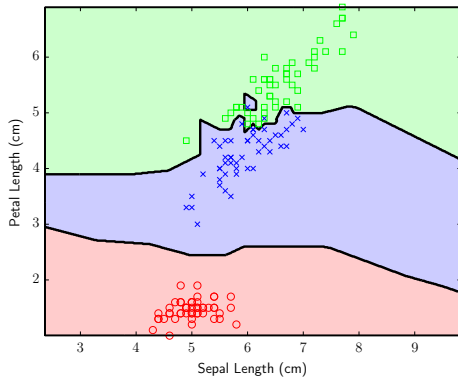
$$\text{mean}_j = \frac{1}{m} \sum_i x_{i,j}$$

$$\text{std}_j = \sqrt{\frac{1}{m} \sum_i (x_{i,j} - \text{mean}_j)^2}$$

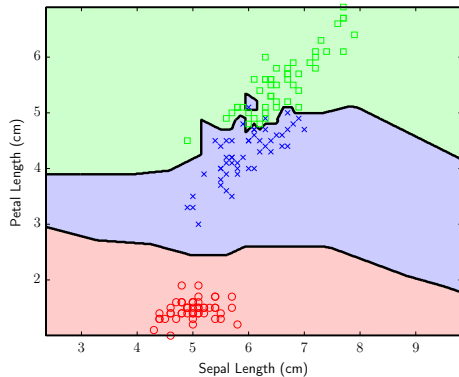
$$x_{i,j} \leftarrow \frac{x_{i,j} - \text{mean}_j}{\text{std}_j}$$

Iris Scaling

“as given” scale

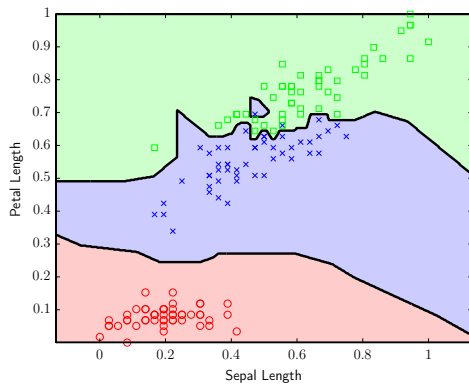


“as given” scale

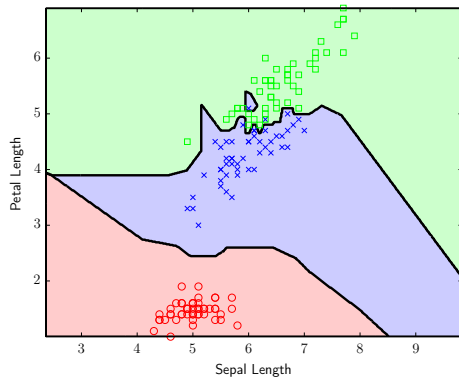


Iris Scaling

range normalized

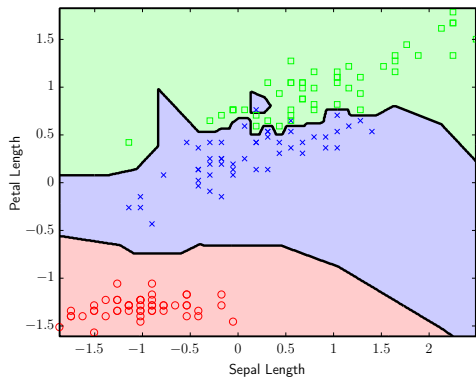


in original axes



Iris Scaling

z-normalized



in original axes

